Project Proposal: Interactive Financial Literacy Platform

1. Problem Statement and Understanding of the Problem

The project aims to make financial education practical, engaging, and accessible to students in Grades 11 and 12, particularly those from underserved communities. Instead of learning only through explanations, students will practise making financial decisions, observe their consequences, and learn how to handle similar situations in real life.

2. Idea or Abstract

We propose a browser-based, game-based financial literacy platform where students learn by navigating realistic situations involving money, education, family responsibilities, and digital purchases.

The game will feature three distinct storylines, each containing five stages, for a total of 15 stages. Students will encounter financial challenges, make choices, manage limited resources, and work towards specific goals.

The learning experience will cover four pillars:

- Plan and Budget: managing expenses, prioritizing needs, and setting goals.
- Save and Manage: building saving habits and preparing for unexpected expenses.
- Understand Investing: learning about risk, compounding, time, and diversification.
- Stay Financially Safe: protecting personal information and recognizing scams.

Each decision will offer four options: Very Bad, Good, Better, and Best. Students will receive feedback explaining the consequences of their choices. If their decisions lead to failure, a recovery mode will explain what went wrong, provide relevant learning resources, and require them to replay that stage before progressing.

The platform will support English, Hindi, and Telugu. At the end of each storyline, students will receive a recap connecting the concepts they learned to practical, real-life situations.

A key feature is student decision tracking. The platform will record choices, mistakes, retries, and progress. This information can help authorized educational organizations identify common learning gaps, provide additional support, and improve the game over time.

3. High-Level Approach, Pipeline, or Architecture

The platform will be developed as a responsive website using React and TypeScript. A predefined game engine(ex.Gemini API) will calculate the consequences of decisions and manage student progress.

An AI service may generate simple, context-specific explanations of mistakes, while the game engine will remain responsible for all outcomes. Optional educational videos will provide further guidance.

A backend and database will store student progress and decision histories. An authorized reporting interface will summarize participation, performance across the four pillars, recurring mistakes, and improvement across attempts. Student information will be protected through restricted access and appropriate data-handling practices.

Development will begin with one complete stage, followed by the remaining stages, language support, and reporting features.

4. Planned Testing Methodology

Testing will verify that decisions produce the correct outcomes, financial calculations are accurate, and students can replay failed stages without losing previous progress. AI explanations will be checked for accuracy, and fallback explanations will be tested.

The English, Hindi, and Telugu versions will be reviewed for clarity and consistency. The platform will also be tested across common browsers and devices, and student records will be checked for accuracy and security.

Finally, pre-game and post-game assessments will measure changes in students' understanding of budgeting, saving, investing, and financial safety. Student feedback and learning reports will guide further improvements.